



Vichara Tharkana

Software Engineer (Undergraduate) | Java & Spring Boot Developer | Web & Android Developer | Cloud & Network Engineering Enthusiast

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Summary

As a Software Engineering undergraduate, I am passionate about building practical and efficient software solutions that solve real-world problems. I enjoy working with modern technologies such as Java, Spring Boot, and web and Android development. I focus on writing clean, structured code and continuously improving my technical skills. I value learning, collaboration, and creating meaningful software that delivers real impact.

Education

Open University of Sri Lanka	2021 – Present
Bachelor of Software Engineering (Honours) GPA-3.04	
St. Joseph's College, Bandarawela	2017
Physical Science, Passed with 3S	

Projects

Gemini-Powered E-Commerce Customer Support Chatbot	2025
• Developed an AI-powered chatbot for e-commerce customer support to provide intelligent and automated assistance for users. • Implemented Retrieval-Augmented Generation (RAG) using Astra DB to improve response accuracy using external knowledge sources. • Enabled smart responses for order tracking, shipping updates, and product-related queries with high contextual relevance. • Integrated conversation memory to maintain context across multiple user interactions for a more natural chat experience.	

Tech Stack- Next.js| TypeScript| Google Gemini API Key| Astra DB

[GitHub](#)

Explore City Android Application	2023
• Built an interactive mobile application designed to help users explore cities with an intuitive and user-friendly interface. • Integrated compass and Google Maps navigation features to provide accurate direction and real-time location guidance. • Enhanced overall user experience by enabling smooth, location-based exploration and easy access to nearby places.	

Vehicle Care System (Web Application)

2024

- Developed a web-based platform for vehicle maintenance services to streamline service booking and management processes.
- Implemented database integration to efficiently manage service records, user data, and maintenance history.
- Designed user interaction features to improve accessibility and ensure smooth communication between users and service providers.

Tech Stack- HTML| CSS| JavaScript| PHP| MySQL
GitHub

Chronic Kidney Disease Prediction System

2025

- Developed a Neural Network-based system to predict Chronic Kidney Disease using patient clinical data.
- Performed data preprocessing, including missing value handling, categorical encoding, and feature scaling.
- Built and trained a TensorFlow/Keras model, achieving **98.75% test accuracy**.
- Designed an interactive prediction interface using Gradio for real-time disease prediction.
- Visualized training accuracy and loss curves for model performance evaluation.

Tech Stack- Python| TensorFlow| Keras| Scikit-learn| Gradio
GitHub

IoT Soil Health Monitoring System

2024

- Designed and developed an IoT-based system for real-time soil health monitoring to support smart agricultural decision-making.
- Collected sensor data from IoT devices and transmitted it to a web dashboard for real-time tracking and analysis.
- Implemented data visualization features to display environmental conditions such as soil moisture and other key parameters for agricultural use.

Tech Stack- Express.js | JavaScript | HTML | CSS | Arduino

Skills

Technical Skills

Java, Spring Boot, MySQL, SQLite, HTML, CSS, JavaScript, PHP, React, Android Studio, Git

Soft Skills

Problem Solving, Teamwork, Leadership, Communication, Creativity, Adaptability

Languages

English, Sinhala

Certifications

Advanced Java Certificate

2021

University of Sabaragamuwa

Windows Server & Hybrid Cloud Administratio (Azure + Microsoft 365)

2026 (Ongoing)

Next Gen Campus